



Quantitative Biophysics and Ecological Dynamics

Bio-Energetics, Native TikZ Trophic Pyramids, and Enzymatic Kinetics

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August 28, 2026

Biophysical Curriculum & Ecological Scope

CURRICULUM

- **1. Bio-Energetic Laws:** Lindeman efficiency and solar assimilation [1].
- **2. Trophic Systems:** Energy tier pyramids and thermodynamic loss.
- **3. Food Web Cascades:** Top-down vs bottom-up regulation [2].
- **4. Enzymatic Kinetics:** Michaelis-Menten and Hill cooperativity [3].
- **5. Population Dynamics:** Lotka-Volterra predator-prey systems.

Core Principle: Energy transfer across biological hierarchies obeys strict thermodynamic conservation laws.

EFFICIENCY

$$\lambda \approx 10\%$$

Trophic Transfer

HEAT DISSIPATION

90%

Metabolic Loss

Ecological Energy Flow: Lindeman's 10% Law

THERMODYNAMICS

1. Lindeman Trophic Efficiency Rule:

$$E_{n+1} = \lambda \cdot E_n, \quad \text{where } \lambda \approx 0.10$$

2. Bio-Energetic Budget Equation:

$$\frac{dB}{dt} = \text{Assimilation} - \text{Respiration} - \text{Excretion}$$

- **Gross Primary Productivity (GPP):** Total solar energy captured.
- **Net Primary Productivity (NPP):**
$$\text{NPP} = \text{GPP} - R_{\text{autotroph}}$$

TROPHIC CASCADE

Tier Biomass Quantities:

- **Apex Predators (Tier 4):** 10 J (Hyper-carnivores).
- **Secondary Consumers (Tier 3):** 100 J (Carnivores).
- **Primary Consumers (Tier 2):** 1,000 J (Herbivores).
- **Primary Producers (Tier 1):** 10,000 J (Plants).

Limitation: The rapid 90% thermodynamic dissipation bounds food chain lengths to 4–5 trophic tiers.

Native TikZ Ecological Trophic Energy Pyramid

ENERGY PYRAMID

- **Solar Input:** 10^6 J incident radiation.
- **Producer Capture:** 10,000 J (1% photosynthetic efficiency).
- **Herbivore Tier:** 1,000 J assimilated.
- **Carnivore Tier:** 100 J transferred.
- **Apex Predator:** 10 J retained.
- **Dissipation:** 90% of absorbed energy radiates as metabolic heat.

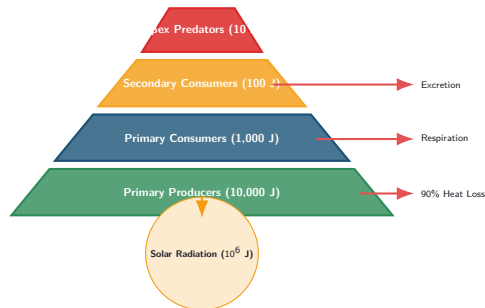


Fig 1: Scalable vector trophic energy tier model.

Enzymatic Kinetics: Michaelis-Menten & Hill Cooperativity

HILL COOPERATIVITY

Cooperative substrate-ligand binding velocity:

$$v = \frac{V_{\max}[S]^h}{K_{0.5}^h + [S]^h}$$

- **Hill Coefficient (h):**
 - $h > 1$: Positive cooperativity (Sigmoidal curve).
 - $h = 1$: Standard Michaelis-Menten (Hyperbolic).
 - $h < 1$: Negative cooperativity.
- **Half-Saturation ($K_{0.5}$):** Concentration at $v = \frac{1}{2} V_{\max}$.

POPULATION DYNAMICS

Lotka-Volterra Predator-Prey Coupled ODEs:

$$\frac{dN}{dt} = rN - aNP \quad (\text{Prey})$$

$$\frac{dP}{dt} = baNP - mP \quad (\text{Predators})$$

Ecological Equilibrium: Non-trivial fixed point occurs at $(N^*, P^*) = \left(\frac{m}{ba}, \frac{r}{a}\right)$.

Enzymatic Parameters & Catalytic Efficiency

Enzyme System	K_m (μM)	k_{cat} (s^{-1})	k_{cat}/K_m ($\text{M}^{-1}\text{s}^{-1}$)
Carbonic Anhydrase	12,000	1,000,000	8.3×10^7
Acetylcholinesterase	95	14,000	1.5×10^8
Triose Phosphate Isomerase	470	4,300	9.1×10^6
Catalase	25,000	40,000,000	1.6×10^9

Catalytic efficiency approaching the diffusion-controlled limit (10^8 – $10^9 \text{ M}^{-1}\text{s}^{-1}$).

KINETIC METRICS

- **Turnover Number (k_{cat}):** Maximum substrate molecules converted per active site per second.
- **Specificity Constant (k_{cat}/K_m):** Measure of catalytic perfection.
- **Diffusion Limit:** Upper bound imposed by Brownian molecular collisions.

Numerical Simulation of Lotka-Volterra Systems

```
1 import numpy as np
2 from scipy.integrate import odeint
3
4 def lotka_volterra_dynamics(state: list, t: float, r: float, a: float, b: float, m: float):
5     """Computes time derivatives for coupled predator-prey populations."""
6     N, P = state
7     dN_dt = r * N - a * N * P
8     dP_dt = b * a * N * P - m * P
9     return [dN_dt, dP_dt]
10
11 # Integrate trajectory
12 t_span = np.linspace(0, 50, 1000)
13 initial_state = [40.0, 9.0]
14 params = (0.5, 0.02, 0.1, 0.2)
15 trajectory = odeint(lotka_volterra_dynamics, initial_state, t_span, args=params)
```

Key Biophysical Takeaways

SUMMARY

- **Trophic Tiers:** Thermodynamic loss constrains ecosystem depth.
- **Enzyme Kinetics:** Hill cooperativity models non-linear allosteric transitions.
- **Dynamic Stability:** Predator-prey phase orbits oscillate around center equilibria.

NEXT STEPS

- **Lab Session 4:** Measure alkaline phosphatase K_m and $V_{\max}$.
- **Problem Set 3:** Derive Lineweaver-Burk linear transformations.
- **Office Hours:** Monday 15:00–17:00 (Bio Lab 108).

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References i

- [1] Raymond L. Lindeman. “**The Trophic-Dynamic Aspect of Ecology**”. In: *Ecology* 23.4 (1942), pp. 399–417.
- [2] Michael Begon et al. ***Ecology: From Individuals to Ecosystems***. Blackwell Publishing, 2006.
- [3] James Keener and James Sneyd. ***Mathematical Physiology: I: Cellular Physiology***. Springer, 2009.